

An Infinite Family of Integer Solutions to $z^2 + y^2z + 2x^3 + 1 = 0$

Abstract

We give an explicit polynomial family of integer solutions to $z^2 + y^2z + 2x^3 + 1 = 0$. The proof is a direct substitution. We then briefly explain the sequence of ansatzes from which the family was found.

1 The family

Theorem 1. *For every odd integer $r \geq 3$, put*

$$\begin{aligned}x_r &= -\frac{r^8 - 8r^6 + 22r^4 - 22r^2 + 5}{2}, \\y_r &= r^3 - 3r, \\z_r &= \frac{r^{12} - 12r^{10} + 57r^8 - 134r^6 + 159r^4 - 84r^2 + 11}{2}.\end{aligned}$$

Then $(x_r, y_r, z_r) \in \mathbb{Z}^3$ and

$$z_r^2 + y_r^2 z_r + 2x_r^3 + 1 = 0.$$

Consequently, the equation has infinitely many integer solutions.

Proof. Set

$$\begin{aligned}a &= r^2 - 2, & n &= \frac{a^2 - 1}{2}, & T &= \frac{r(a - 1)}{2}, \\X &= 2n^2 + a, & M &= 4n^3 + 3an + 1.\end{aligned}$$

Since r and a are odd, all five quantities are integers. Direct expansion gives

$$-X = x_r, \quad 2T = y_r, \quad M - 2T^2 = z_r,$$

and

$$M^2 - 2X^3 + 1 = \frac{(a - 1)^4(a + 2)^2}{4} = 4 \left(\frac{r(a - 1)}{2} \right)^4 = 4T^4.$$

Therefore, with $(x, y, z) = (-X, 2T, M - 2T^2)$,

$$\begin{aligned}z^2 + y^2z + 2x^3 + 1 &= z(z + y^2) - 2X^3 + 1 \\&= (M - 2T^2)(M + 2T^2) - 2X^3 + 1 \\&= M^2 - 2X^3 + 1 - 4T^4 = 0.\end{aligned}$$

Finally, $y_r = r^3 - 3r$ is strictly increasing for $r \geq 3$, so the solutions obtained from distinct odd values of r are distinct. $\square$

2 How the construction was found

This section records the search that led to the formulas; it is not an additional proof.

Write $x = -X$. The equation becomes

$$z(z + y^2) = 2X^3 - 1.$$

Thus the aim is to factor $2X^3 - 1$ into two integers whose difference is a square. Taking $y = 2T$ and centering the factors symmetrically,

$$z = M - 2T^2, \quad z + y^2 = M + 2T^2,$$

reduces the problem to

$$M^2 - 2X^3 + 1 = 4T^4. \tag{1}$$

There is an obvious polynomial solution of the unperturbed relation $M^2 = 2X^3$:

$$X = 2n^2, \quad M = 4n^3.$$

This suggests perturbing it by writing

$$X = 2n^2 + a, \quad M = 4n^3 + bn + c.$$

In the expansion of $M^2 - 2X^3 + 1$, the coefficient of n^4 is $8(b - 3a)$. Choosing $b = 3a$ removes this highest-order error, and the simple choice $c = 1$ gives

$$X = 2n^2 + a, \quad M = 4n^3 + 3an + 1,$$

with residual

$$M^2 - 2X^3 + 1 = 8n^3 - 3a^2n^2 + 6an - 2a^3 + 2.$$

The leading terms indicate that n should be approximately $a^2/2$. Accordingly, one tries

$$n = \frac{a^2 + pa + q}{2}$$

and asks that four times the residual be the square of a monic cubic in a . Elementary coefficient matching gives the particularly simple choice

$$p = 0, \quad q = -1,$$

so that

$$n = \frac{a^2 - 1}{2}$$

and

$$\begin{aligned} 4(M^2 - 2X^3 + 1) &= (a^3 - 3a + 2)^2 \\ &= (a - 1)^4(a + 2)^2. \end{aligned}$$

The factor $(a - 1)^4$ is already a fourth power. To make the remaining factor a fourth power as well, impose

$$a + 2 = r^2, \quad \text{that is,} \quad a = r^2 - 2.$$

Then

$$M^2 - 2X^3 + 1 = 4 \left(\frac{r(a - 1)}{2} \right)^4,$$

which is exactly (1) with $T = r(a - 1)/2$. Reversing the substitutions produces the family in Theorem 1. The large polynomials in r arise only from eliminating the intermediate variables a, n, T, X, M .